

Bhavya Shah

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EXPERIENCE

Summer Intern — Ford Motor Company

June 2026 – July 2026

Sanand Engine Assembly Plant (SaEP), Sanand, Gujarat, India

2 Months

- Developed **AI in MOS**, a forecasting platform integrating machine and material modules within a single interface.
- Built a **Machine Forecasting Module** using time-series models on machine data to enable proactive predictive maintenance.
- Created a **Material Module** analyzing spare part consumption, BOM, and lead times to optimize procurement and reduce stock costs.

EDUCATION

Nirma University, Ahmedabad

2023–2027

B.Tech in Computer Science and Engineering

CGPA: 8.71

SKILLS

Technical Skills: C/C++, JavaScript, Node.js, Next.js, Python3, MySQL

Coursework: Deep Learning, Operating Systems, Database Management System, Object-Oriented Programming, Computer Networks, Computer Architecture, Electronic Systems Engineering, Cloud Computing

PROJECTS

RiskAnalyzer: Alternative Data Credit Risk Model

Python, Flask, React.js

Developed a machine learning model and full-stack web application to predict credit default probability using non-traditional data sources (e.g., mobile usage, utility payments), enabling inclusive risk assessments for ‘thin-file’ customers.

[Demo Link](#) [GitHub Link](#)

ResQNet

React.js, Node.js, FastAPI, Python, Blockchain, LLM

Built a decentralized, reward-based SOS platform featuring ML-based emergency classification and automated volunteer dispatch via Dijkstra routing. Deployed Ethereum smart contracts (Solidity, Hardhat) on Sepolia testnet with a 3-step verification system for automated ETH reward distribution. Engineered an integrated fraud-prevention system to eliminate fake alerts and maintain platform integrity.

[Demo Link](#) [GitHub Link](#)

VoxelWorld

JavaScript, WebGL, Neural Network

Lightweight, browser-based 3D voxel sandbox featuring procedural environments, dynamic time-of-day lighting, and a custom built-in Neural Network for generating instant architectural structures.

[Demo Link](#) [GitHub Link](#)

ACHIEVEMENTS

- **Winner**, DotSlash 9.0 (SVNIT, Surat) - 1st place among 800+ teams with ResQNet
- **Winner**, ODOO x KSV Hackathon 2026 - 1st place among 836 teams with CoRYD
- **2nd Runner Up**, Hack-a-Mind (Nirma University & SUNY Binghamton University)
- **Finalist**, FinShield Hackathon 2025 (IIT Hyderabad)

RESEARCH WORK

TinyML-Based Trojan Detection Framework in Smart Grid Systems

Published, 15th SGC 2025

Proposed a TinyML-driven intrusion detection model for smart grids achieving 98.39% accuracy with 87% storage compression, demonstrating efficient, real-time, edge-based cybersecurity.

TinyML-Based Efficient Framework for Industrial IoT Communication Using 5G Network Slicing

Published, SpliTech 2026

Designed a scalable 5G network slicing system with TinyML quantized models (LinearSVM, DecisionTree, ANN), improving latency and efficiency while maintaining 82.1% accuracy with 77% smaller model size.

Machine and Reinforcement Learning-Based Collaborative Framework for Drone Navigation in Battlefield Environment

Preprint

Developed a Q-learning-based reinforcement learning framework integrated with ML classifiers for autonomous UAV navigation, achieving enhanced adaptability and collision avoidance in dynamic battlefield environments.

POSITION OF RESPONSIBILITY

Technical Director, ACES (Association of Computer Engineering Students) - Lead organizer of technical events including Insignia Hackathon, Summer Internship Talks, and various other student-facing programs.